

INTERNATIONAL GCSE **BIOLOGY**

9201/1

Paper 1

Mark Scheme

November 2024

Version: 1.0 Final



2 4 B Y 9 2 0 1 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	glucose oxygen		1 1	AO1 3.2.6 b, c, d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	any two from: • movement or muscle contraction • maintain a steady body temperature • build large molecules (from smaller ones) • active transport	allow for exercise allow to keep warm allow build proteins allow moving substances against concentration gradient	2	AO1 3.2.6 f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	stopwatch / stopclock	allow clock / watch with second hand ignore timer / phone unqualified ignore clock unqualified	1	AO4 3.2.6 g 6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	any two from: • exercise for same / specific length of time • control type of exercise • use more participants to calculate mean or repeat test multiple times to calculate mean or repeat to identify and ignore anomalies • shorten the time interval for counting • use a pulse / heart rate monitor (to measure heart rate)	allow same intensity of exercise	2	AO4 3.2.6 g 6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	random		1	AO4 3.2.6 g 6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	(so) resting pulse / heart rate can be measured		1	AO4 3.2.6g 6.2.4
	to compare pulse / heart rate before and after exercise		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	lactic acid	must be in this order	1	AO1 3.2.6 i, j
	water		1	

Total			12	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	cell membrane	must be in this order	1	AO1 3.1.1 a, b 3.2.6 e
	mitochondrion		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	(A) chloroplast		1	AO1 3.1.1 b
	(B) (permanent) vacuole		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	cell wall, cytoplasm, cell membrane		1	AO3 3.1.1 b, c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	24^3	allow $24 \times 24 \times 24$	1	AO2 3.1.1 b, c 6.3.14
	13 824 (mm ³)	allow 1.3824×10^4	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	$96 \div 64$		1	AO2 3.1.1 b, c 6.3.6
	1.5		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	(so) exchange of all necessary substances can take place (across the cell membrane)	allow for increased diffusion allow absorb nutrients faster	1	AO1 3.1.5 h

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	photosynthesis	allow to produce glucose	1	AO1 3.1.4 a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	leaf		1	AO1 3.1.2 b, c 3.1.4 b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.9	any one from: • muscular • glandular • epithelial	allow phonetic spelling allow muscle allow epithelium	1	AO1 3.1.3 b

Total			13	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	(A) right atrium	in this order only	1	AO2 3.2.3 c
	(B) left ventricle	allow 1 mark if atrium and ventricle correct but right and left interchanged	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	vein	allow vena cava allow pulmonary vein	1	AO1 3.2.3 i

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	right atrium		1	AO1 3.2.3 d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	oxygen	must be in this order	1	AO1 3.2.3 m, n 3.2.4 a
	carbon dioxide		1	
	glucose		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	a stent		1	AO1 3.2.3 e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	reduced / no blood flow (to heart)		1	AO2 6-7 3.2.3 e
	(so) less / no oxygen (to heart muscle cells) for respiration or (so) less / no glucose (to heart muscle cells) for respiration or less / no energy released from respiration	do not accept energy needed for respiration	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	arteries have thick walls	allow arteries have thick elastic / muscle tissue	1	AO2 6-7 3.2.3 j
	to stop artery from bursting (at high pressure)	allow elastic tissue allows stretching with the (force of the) pulse / heart beat	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	thin walls	allow wall one cell thick do not accept cell wall	1	AO1 3.2.3 k 3.1.5 i
	(thin walls) to provide a short diffusion pathway		1	

Total		14
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	nucleus		1	AO1 3.5.2 a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	one form of a gene		1	AO1 3.5.2 b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	four gametes		1	AO1 3.5.2 d, e, g, h

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	any two from: • for growth • for asexual reproduction • so old / damaged cells / tissues can be replaced	allow (so) tissues can be repaired do not accept to repair cells	2	AO1 3.5.2 d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	most animal cells differentiate at an early stage of development but many plant cells differentiate throughout life		1	AO2 3.5.2 i, j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	19		1	AO2 3.5.2 c, h 6.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.7	bone marrow	allow other named sources eg eyes, brain or skin	1	AO1 3.5.2 k

Question	Answers	Mark	AO/ Spec. Ref
04.8	Level 2: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	3–4	AO3 3.5.2 k, l, m
	Level 1: Some logically linked reasons are given. There may also be a simple judgement.	1–2	
	No relevant content	0	
	Indicative content Advantages of using embryonic stem cell - different types of cells can be produced to replace the various types of cells that make up the lung - easier to manipulate and grow embryonic stem cells than adult stem cells - less likely to contain environmental toxins Disadvantages of using embryonic stem cell - embryo cannot give consent - embryo is killed / destroyed by the procedure - possible risk of rejection by the patient's immune system - drugs that suppress the immune system may be needed - more likely that uncontrolled / faster cell division could lead to a tumour or cancer For level 2 an advantage and disadvantage of using embryonic stem cells are given. Allow converse statements that clearly relate to adult stem cells instead of embryonic stem cells.		

Total		12
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	spinal cord		1	AO1 3.4.1 b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	to coordinate responses (to stimuli or changes in the environment) or to coordinate behaviour		1	AO1 3.4.1 a, b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	receptors (in the skin / sense organs) detect the stimulus / heat impulses (from the receptors) travel along the sensory neurone to the CNS / spinal cord impulses travel from relay to motor neurone (impulse travels to) the effector / muscle contracts (and the hand is pulled away)	max 3 marks if neurone sequence incorrect no credit for letters unless named if no other marks awarded, allow 1 mark for 3 or 4 letters correctly named	1 1 1 1	AO1 3.4.1 a, c, d, e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	neurone 1 releases chemical	allow neurotransmitter for chemical	1	AO3 AO2 3.4.1 d
	chemical diffuses / passes to neurone 2		1	
	(chemical) causes an impulse in neurone 2		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	drug A prevents the release of the chemical at the synapse	allow drug A stops the chemical binding to second neurone	1	AO3 3.4.1 d
	(so) pain impulse is not caused in another neurone or (therefore) pain impulse does not reach the brain	allow drug A destroys chemical released at synapse	1	

Total		11
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	guard (cells)		1	AO2 3.2.2 e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	(in hot dry environments) leaves lose water faster by evaporation / transpiration	allow (in hot dry environments rate of) evaporation / transpiration increases	1	AO1 3.2.2 d
	(but) water cannot be replaced quickly enough by the roots		1	
	(so) stomata close to prevent water loss which would lead to wilting		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	(conversion from cm to mm) $8 \times 10 = 80 \text{ mm}$	allow conversion at any point	1	AO2 3.2.2 b, c, d 6.3.14
	(radius = diameter divide by 2) $80 \div 2 = 40$		1	
	(area) $3.14 \times 40 \times 40$		1	
	5024	allow 5026.55 (using π key on calculator)	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	$(45 + 47) \div 2$		1	AO2 3.2.2 d 6.3.7
	46	allow $(45 + 20 + 47) \div 3$ = 37(.333) for 1 mark	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	(rose)			AO3 3.2.2 c, d
	no stomata on upper surface to reduce water loss by evaporation		1	
	because upper surface is warmer / hotter or because upper surface is in direct sunlight	allow it is cooler / moister underneath leaf (so less water loss)	1	
	(water lily)			
	lower surface is under water so no stomata to prevent water entering	if neither mark awarded allow 1 mark for lower surface is under water so no gas exchange can take place	1	
	(therefore) stomata on upper surface needed for gaseous exchange		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.6	(blue) water moved upwards (in the stem)	allow (blue) water moved upwards (to the leaves)	1	AO2 3.2.2 f
	through the xylem to the leaves	allow through the xylem in the stem	1	
	blue food colour left in leaves when water evaporated		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
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06.7	phloem		1	AO1 3.2.2 f
	carries (dissolved) sugar from leaves to rest of the plant	allow sucrose for sugar allow carries (dissolved) sugar from leaves to named part of plant ignore glucose	1	

Total		19
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Question	Answers	Mark	AO/ Spec. Ref
07.1	Level 2: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	3–4	AO2 3.3.3 a, b, c, e 3.3.4 a, d
	Level 1: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	1–2	
	No relevant content	0	
	Indicative content • compost can be used to improve soil fertility / structure • as natural fertiliser adds minerals / ions to soil • to increase yield of vegetables / crops • (so) less need for artificial fertilisers • less need for waste sites • less pollution • (so) less methane production from waste sites • (and) reduced global warming • reduced named effects of global warming • less infestation with flies / rats • less risk of transmission of diseases For level 2 reference to advantages to the gardener / vegetables and to environmental issues		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	carbon dioxide from respiration of microorganisms		1	AO2 3.2.1 d, e 3.3.3 c, e
	(carbon dioxide) is used in photosynthesis to produce glucose		1	
	glucose used to produce starch / fats / oil / cellulose / protein	allow to release energy	1	
	nitrate (ions) released into soil		1	
	nitrate (ions) used to make amino acids / proteins for growth	allow magnesium ions are used to make chlorophyll for photosynthesis / growth	1	
Total			9	